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Functor Categories $[C,D]$: Categories of Functors

Definition

The **functor category** $[C,D]$ is the category whose: - **Objects**: Functors $F: C \rightarrow D$ - **Morphisms**: Natural transformations between functors

Core Axioms

$[C,D]$ is a legitimate category:

1. **Identity**: For each functor F , identity natural transformation $\text{id}_F: F \rightarrow F$ ($\text{id}_F(X) = \text{id}_{\{F(X)\}}$)
2. **Composition**: For $\eta: F \rightarrow G$ and $\theta: G \rightarrow H$, their composition $(\theta \circ \eta): F \rightarrow H$ is defined componentwise:

$$(\theta \circ \eta)_X = \theta_X \circ \eta_X: F(X) \rightarrow H(X)$$
3. **Associativity**: $(\theta \circ \eta) \circ \phi = \theta \circ (\eta \circ \phi)$
4. **Identity Laws**: $\text{id}_G \circ \eta = \eta$ and $\theta \circ \text{id}_F = \theta$

Key Observation

Morphisms in $[C,D]$ are **natural transformations** composition respects the naturality squares.

8 Critical Examples

1. **$[C^{op}, Set]$ = Presheaf Category**
 - Fundamental in category theory
 - Every category embeds densely here via Yoneda embedding
 - Limits/colimits computed pointwise
2. **Simplicial Sets $[\Delta^{op}, Set]$**
 - Objects: simplicial complexes with degenerate faces

- Used in algebraic topology, higher category theory
 - Quasi-categories live in here
3. **[G, Set] = G-Sets (Group Actions)**
- Objects: functors from one-object-one-morphism G to Set (= G -actions)
 - Morphisms: natural transformations (= equivariant maps)
 - Category of G -sets and G -equivariant functions
4. **[J, Ab] (Direct Limits)**
- Objects: functors $J \rightarrow \text{Ab}$ (families of abelian groups)
 - Colimit in $[J, \text{Ab}]$ computed in Ab gives direct limit
 - The directed limit $\varinjlim$ is the colimit in this functor category
5. **[$\mathcal{A}^{\text{op}}$, Grp] (Inverse Limits)**
- Objects: sequences of groups and homomorphisms
 - Inverse limit is the limit in this functor category
 - $\varprojlim$ computed as product of stabilizing sequences
6. **[2, Set] (Spans and Correspondences)**
- Objects: functors from 2-element category $\{0 \rightarrow 1\}$ to Set
 - = Pairs of functions with common codomain
 - Morphisms: commutative triangles
7. **[$\mathcal{C}$, C] (Simplicial Objects in C)**
- Objects: simplicial objects in any category $\mathcal{C}$
 - Used in homological algebra, derived functors
 - Simplicial complexes, simplicial abelian groups, etc.
8. **[C, $\mathcal{D}$] (Product of Functor Categories)**
- Products of functor categories are functor categories
 - $[C, \mathcal{D} \times \mathcal{E}] \cong [C, \mathcal{D}] \times [C, \mathcal{E}]$
 - Functors into product = pair of functors into factors

Fundamental Properties

Limits and Colimits (Computed Pointwise)

Theorem: Limits and colimits in $[C, D]$ are computed **pointwise**:

If $\cdot : C \rightarrow D$ and we want the limit, for each object X in C :

$$(\lim \cdot)_X = \lim \cdot_X$$

Why: Natural transformations are compatible with limits, so we can compute limits of components independently.

Example: - $\lim F$ in $[C, \text{Set}]$ is the functor that assigns to each X the limit $\lim_{j \in J} F(j)(X)$ - Yoneda embedding $y: C \rightarrow [C^{\text{op}}, \text{Set}]$ preserves limits

Completeness and Cocompleteness

Theorem: If D is complete/cocomplete, so is $[C, D]$.

Proof: Use pointwise computation of limits/colimits.

Significance: This means: - $[C, \text{Set}]$ is complete and cocomplete - Presheaves have all limits and colimits - Every category embeds into a complete/cocomplete category

Yoneda Embedding: $C \rightarrow [C^{op}, Set]$

The Yoneda embedding is the **fundamental functor**:

$y: C \rightarrow [C^{op}, Set]$
 $X \mapsto Hom(-, X)$
 $(f: X \rightarrow Y) \mapsto (\text{natural transformation: post-composition by } f)$

Theorem (Yoneda Fully Faithful): y is fully faithful. - Injective on objects (up to isomorphism) - Bijective on morphisms

Theorem (Yoneda Density): Every presheaf is the **colimit of representables**.

$F: [C^{op}, Set] = \text{colim}_{\{X(\wedge F)\}} Hom(-, X)$

where $(\wedge F)$ is the comma category of representables under F .

Fundamental Theorem of Category Theory

Theorem: Every functor $F: C \rightarrow D$ can be **uniquely extended** to a continuous functor (preserving limits) from $[C^{op}, Set]$ to $[D^{op}, Set]$:

$$\begin{array}{ccc} C & \xrightarrow{F} & D \\ | & & | \\ y & & y \\ [C^{op}, Set] & \xrightarrow{\text{Ran}_y F} & [D^{op}, Set] \end{array}$$

This extension is given by the **right Kan extension** of F along the Yoneda embedding.

Relation to Kan Extensions

Everything in functor categories is built from Kan extensions:

1. **Yoneda embedding:** $y_C: C \rightarrow [C^{op}, Set]$ is defined by Kan extensions
2. **Limits in $[C, D]$:** Computed via Kan extensions in D
3. **Extensions of functors:** Via Kan extensions along Yoneda
4. **Density theorem:** Every presheaf is colimit of representables (= left Kan extension)

Study Checklist

- ☐ Understand $[C^{op}, Set]$ as the free cocompletion of C
- ☐ Recognize $[C^{op}, Set]$ in multiple examples
- ☐ Verify Yoneda lemma in simple cases
- ☐ Compute limits/colimits pointwise in $[C, D]$
- ☐ See connection to Kan extensions

Key Insight

Functor categories are the bridge between individual categories and the universal properties that define them. In $[C, D]$, natural transformations are the **morphisms**, which means **Kan extensions are the universal morphisms** in functor categories.

This is why $[C^{\text{op}}, \text{Set}]$ is so important: it's the **universal** category containing C (via Yoneda), and every construction in category theory can be understood as happening in some functor category.